

Chemnitz University of Technology's solar cell printed on paper

of paper. Special inks with electrical properties form the necessary structures on paper, which ensure that electricity is generated when being exposed to light. Since the employed conventional printing methods are very cost-efficient, the printed solar panels will generate much cheaper electricity than conventional solar cells.

Prof. Dr Arved Hübner from the Institute for Print and Media Technology at Chemnitz University of Technology, who is working together with his research team on the 3PV technology for more than three years now, speaks of a paradigm shift in solar technology. His vision for the future is that common printing houses around the world could produce and

market 3PV solar panels.

A printing speed of up to 15 m/s is possible. Besides, 2D patterning is also possible.

Dye-sensitised solar cell to enhance power efficiency

Researchers at Northwestern University have developed a stable dye-sensitised solar cell that may turn out to be cheaper than silicon-based cells. The all-new Grätzel cell design, named after the Swiss chemist, replaces silicon with titanium-oxide semiconductor. This is much cheaper, readily available and less toxic, thereby improving solar energy's green credentials.

However, there is one challenge with the new design: Its electrolyte is

made of an organic liquid that leaks, threatening to corrode the cell. This problem has been resolved by the researchers with the use of a new material that solidifies, thus avoiding leaks and making the solid-state cell more stable. The material used is a thin-film compound called CsSnI_3 , — a combination of cesium, tin and iodine. It is added in liquid form to dye-coated nanoparticles and then it turns into a solid mass, Gizmag reports.

According to the lead researcher involved in the project, this is the first demo of a solidstate dye-sensitised solar cell that can definitely ensure better efficiency than the current solar technologies.

Magazines to embed videos, interactive button

For years the debate about the future of print medium is on, especially with the growing popularity and immense reach of the Internet. The worries of print publishers are, however, going to reduce with the launch of an interesting application.

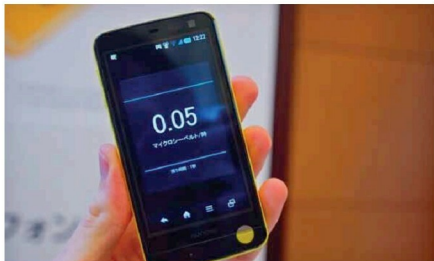
Software developer Layar recently unveiled an application called the Layar Creator that will help publishers more easily link their print and digital content, hoping to jumpstart the so-called mobile augmented reality business. This means when you open your magazine, you may be able to watch superimposed videos and hottest sale options that pop up on the page by using your mobile camera.

This technology is expected to move into the mainstream in a few years, as chipset vendors will soon start work on enabling features like image recognition in smartphones, tablets and PCs.

Radiation-detecting phone

Imagine a phone that could detect harmful radiations around you. Well, this unique technology has actually been brought to the mobile phones by a Japanese company called Softbank.

Softbank has entered the market with its first Pantone 5 107SH smartphone, which comes with an in-built



Softbank's Pantone 5 107SH smartphone that detects radiations